EDA Plots Explained

All five plots are generated by `src/eda.py` and saved to `reports/figures/`. Open the actual PNG files alongside this file — this document explains what each one shows and, more importantly, *why it matters for the rest of the project*.

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1. `class_balance.png` — Default vs. Non-Default

What it shows: A simple bar chart of how many customers did and didn't default. ~23,300 customers (78%) did not default, ~6,700 (22%) did.

Why it matters: This is the single most important chart in the whole project, because it's the reason three other decisions were made downstream:

- It's why `class_weight="balanced"` is used on every model (see `03_metrics_explained.md`).
- It's why accuracy alone is a misleading metric — a model that always predicts "no default" is already 78% accurate while being useless.
- It's why F1-score (not accuracy) is used to pick the winning model.

How to say it in an interview: *"The first thing I checked was the class balance, because everything else — the evaluation metric, the class weighting, even which model wins — depends on how imbalanced the target is. At 22% positive, this isn't a severe imbalance, but it's enough that raw accuracy would be misleading."*

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2. `default_rate_by_education.png` — Default Rate by Education Level

What it shows: The default rate for each EDUCATION category (1=grad school, 2=university, 3=high school, 4=other). Default rate rises from ~19% (grad school) to ~24% (university) to ~25% (high school), then drops to ~7% for "other."

Why it matters: Two things: 1. It confirms EDUCATION is actually predictive — default rate isn't flat across categories, so it's worth keeping as a model feature. 2. The "other" category's oddly low rate (7%, on a small group) is a mild red flag for a small/noisy bucket — exactly the kind of thing worth noticing during EDA rather than blindly trusting every category.

How to say it in an interview: *"Default rate roughly tracks education level in the direction you'd expect — more advanced degrees, lower default rate — except for the 'other' bucket, which is small and is exactly the group I folded the undocumented category codes (0, 5, 6) into during cleaning. Its low rate is partly a sign that it's a mixed, noisy group, not a sign education doesn't matter."*

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3. `utilization_by_default.png` — Credit Utilization Ratio by Default Status

What it shows: An overlaid histogram of `CREDIT_UTIL_RATIO` (bill amount ÷ credit limit) for defaulters (orange) vs. non-defaulters (blue). Non- defaulters are weighted much more heavily toward low utilization (near 0); defaulters are shifted toward higher utilization, with a visible bump around 0.8–1.0 (i.e., near or at their credit limit).

Why it matters: This is the direct visual justification for engineering `CREDIT_UTIL_RATIO` in the first place. A raw bill amount like "$50,000 owed" means nothing on its own — it's a red flag for someone with a $60,000 limit and completely normal for someone with a $500,000 limit. This plot shows that once you normalize for the credit limit, there's a real, visible separation between the two groups.

How to say it in an interview: *"Raw bill amounts aren't comparable across customers with different credit limits, so I engineered a utilization ratio — bill divided by limit. This plot is what justified that: you can see defaulters cluster much more heavily near full utilization, while non-defaulters cluster near zero. That's a much cleaner signal than the raw dollar amount would give a model."*

(Caveat worth knowing: the correlation heatmap below shows this feature's linear correlation with default is modest — 0.09 — smaller than the visual separation might suggest. That's because the relationship isn't purely linear; trees pick up on it better than linear correlation does. Mentioning this unprompted shows you're not just reading plots at face value.)

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4. `default_rate_by_delayed_payments.png` — Default Rate by Number of Delayed Payments

What it shows: Default rate climbs steadily and almost linearly with `NUM_DELAYED_PAYMENTS`: ~12% at 0 delayed months, up to ~70% at 6 delayed months (i.e., late every single month in the 6-month window).

Why it matters: This is the strongest, cleanest signal in the entire EDA — and it's confirmed numerically in the correlation heatmap (0.40 correlation with default, by far the highest of any engineered feature). It's also the clearest real-world justification in the whole project: repeated recent late payments is exactly the kind of behavior a human underwriter would flag too, so the model's strongest signal lines up with common sense.

How to say it in an interview: *"This was the single most useful chart I produced. Default rate goes from 12% with zero delayed payments to 70% with six — almost a straight line. It's also the feature with the highest correlation to the target, 0.40, well above anything else. If I only had time to build one feature, this would be it."*

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5. `correlation_heatmap.png` — Correlation Heatmap

What it shows: Pairwise correlation between `LIMIT_BAL`, `AGE`, `CREDIT_UTIL_RATIO`, `NUM_DELAYED_PAYMENTS`, and `default`.

Key numbers: - `NUM_DELAYED_PAYMENTS` <-> `default`: **+0.40** (by far the strongest) - `LIMIT_BAL` <-> `default`: **-0.15** (higher credit limit -> lower default rate — makes sense, banks already give higher limits to lower-risk customers) - `CREDIT_UTIL_RATIO` <-> `default`: **+0.09** (positive but modest — see the caveat in section 3) - `AGE` <-> `default`: **~0.01** (essentially no linear relationship)

Why it matters: This is the "sanity check" plot — it turns the visual impressions from the other four plots into one number each, and it's a fast way to explain feature importance before even training a model.

How to say it in an interview: *"The correlation heatmap summarizes everything the other plots showed. Delayed payments is the standout predictor at 0.40. Credit limit has a weaker negative correlation, which makes sense — it's partly a proxy for how risky the bank already judged someone to be when they set the limit. Age barely correlates at all, which is useful to know — it tells me the model isn't going to lean heavily on age, which also matters from a fairness standpoint."*